

TEAM DETAILS

Team Name - Algorithm

Theme - Cybersecurity/DevSecOps (Development + Security + Operations)

Project Name - ReleaseSheild AI

Mentor Name - Mr Gaurav Nayak

College / Institute Name - LNCT Bhopal

PROBLEM STATEMENT

Why small teams are stuck doing security triage by hand

Trivy
dependency scan

Semgrep
code scan

OWASP ZAP
live app scan

Gitleaks
secret detection

no shared pipeline — no correlation, no routing

↓

Manual triage

Someone manually cross-checks every report
Pings a developer based on guesswork
No SLA. No accountability.

Enterprise ASPM
Apiiro · Cycode · Wiz

Weeks of setup
Sales-call gated, enterprise priced
Built for large orgs, not 10-person teams

Single-purpose scanners
Snyk · Semgrep alone

Fixes one layer only
No cross-tool correlation
No ownership routing at all

Small teams are left doing all of this by hand.

THREE CORE PROBLEMS

NO CORRELATION

- Same issue flagged by 2 tools = 2 tickets
  No dedup across scanners
- Manual cross-referencing required
- Wastes engineering time every week

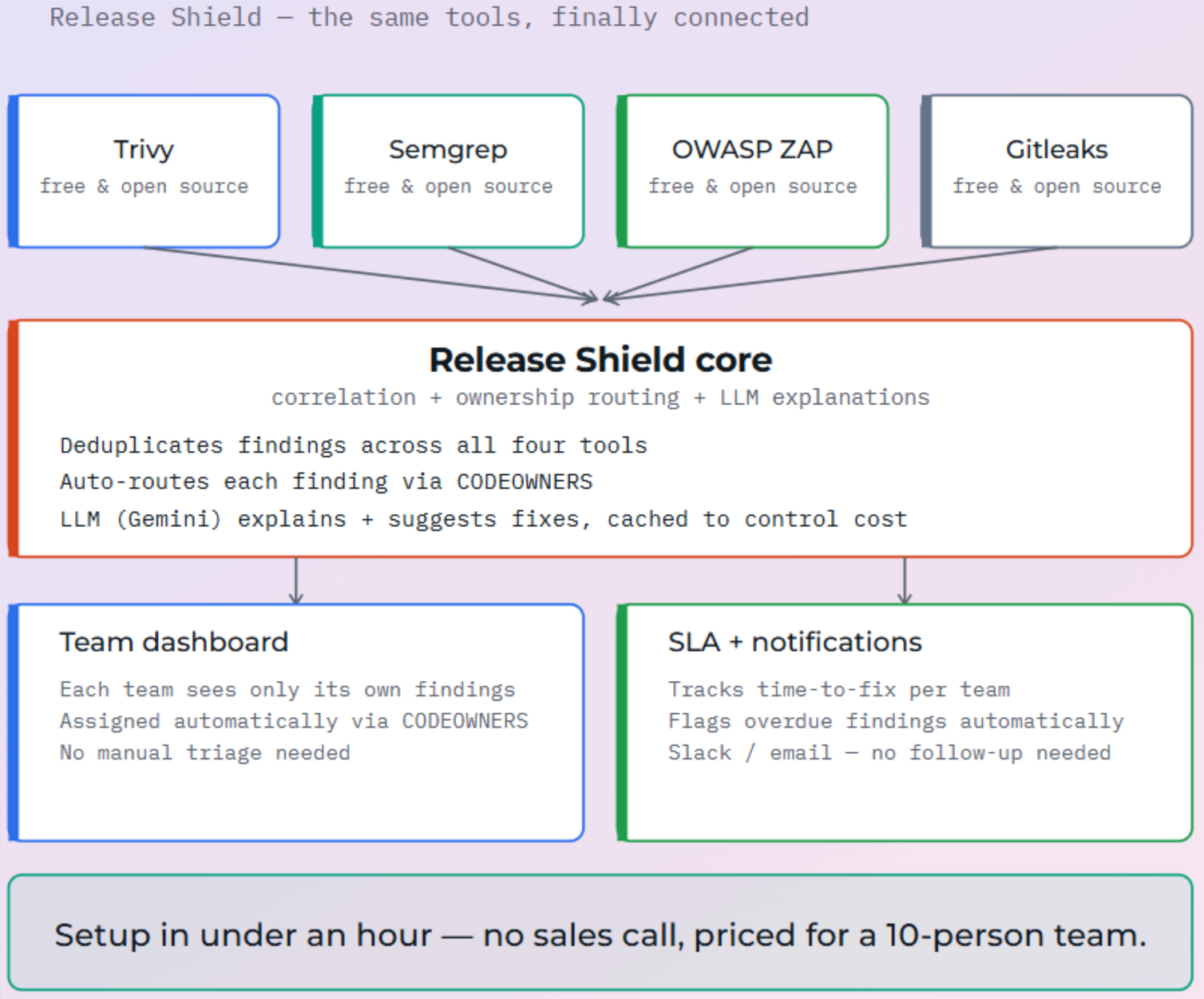
NO OWNERSHIP

Findings have no assigned ownerDeveloper identification is guessworkFixes get lost or delayedNo accountability trail

NOT BUILT FOR SMALL TEAMSENTERPRISE

- Enterprise ASPM = weeks of setupSales-call gated, enterprise pricingSingle scanners solve one layer onlyNothing priced for a 10-person team

PRODUCT DESCRIPTION & PROPOSED SOLUTION



WHY RELEASE SHIELD AI WINS

SPEED

Setup in under an hourSelf-serve — no sales call neededCI/CD from day one (GitHub Actions)Live on your first PR the same day

SIMPLICITY

One dashboard instead of fourCODEOWNERS routing, not commit-history AIPlain-language explanation per findingNo enterprise onboarding process

PRICE

Built on free, open-source scannersLLM cost ~$1–20/month per client, cachedPriced for a 10-person teamNot competing with Apiiro/Wiz on depth — winning on price instead

EXISTING SOLUTIONS

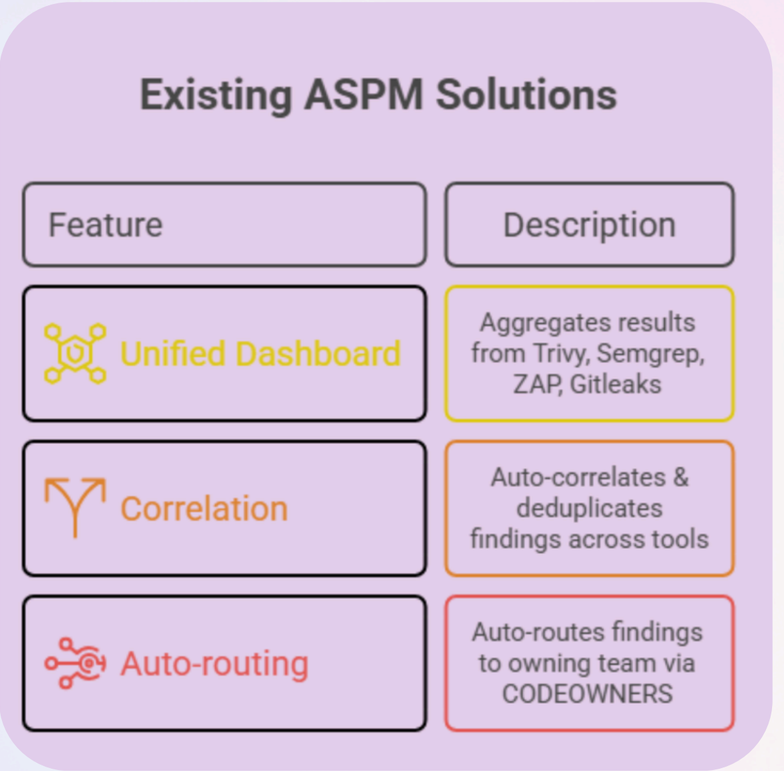
- Unified ASPM (Application Security Posture Management) dashboard aggregating results from Trivy, Semgrep, ZAP, Gitleaks.
- Auto-correlates & deduplicates findings across tools.
- Auto-routes findings to owning team via CODEOWNERS — no manual handoff.

UNIQUENESS

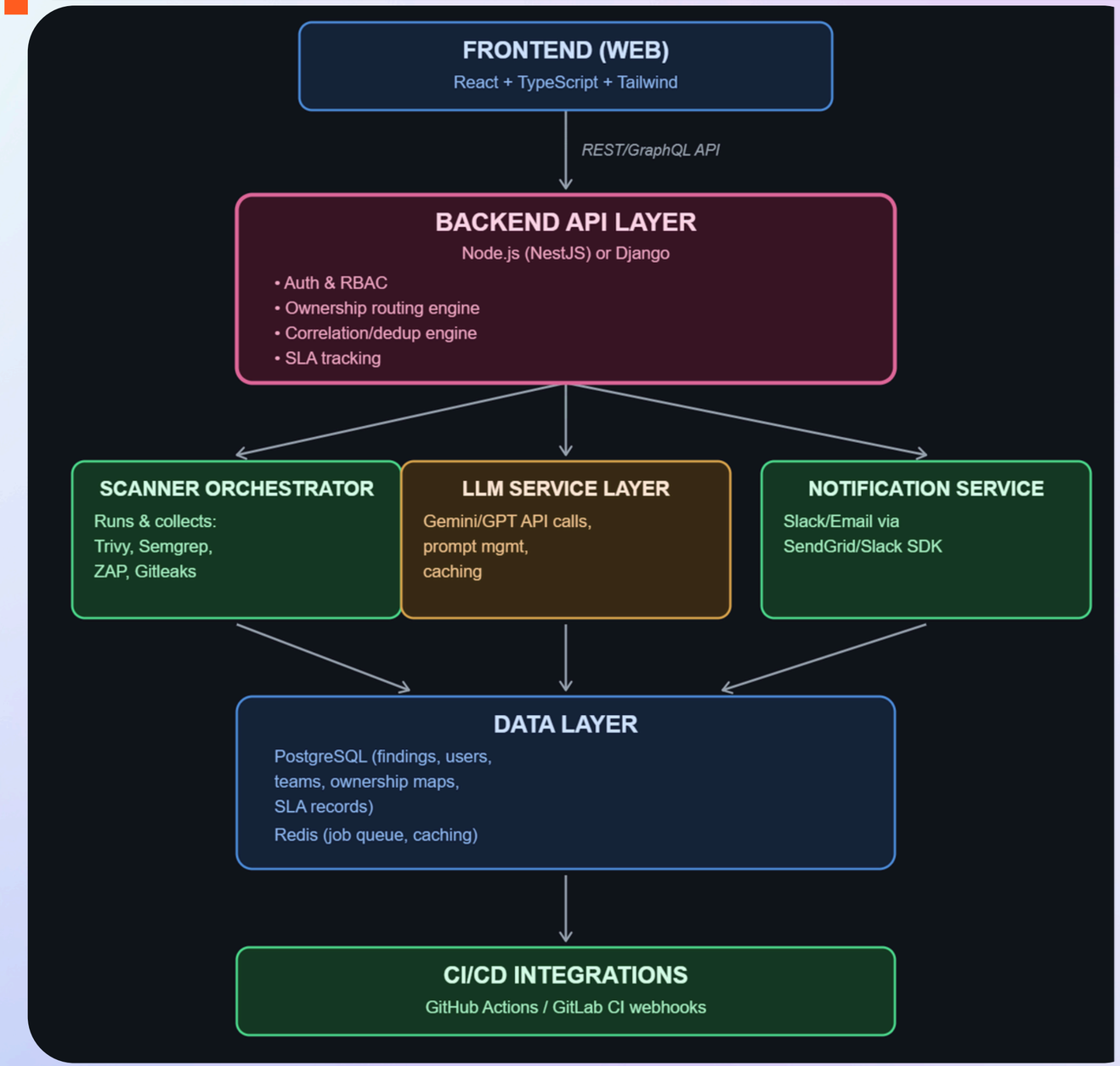
- No new scanning engine — combines existing free tools.
- Automatic ownership routing (not manual).
- Setup in under 1 hour, no sales calls, priced for 10-person teams.

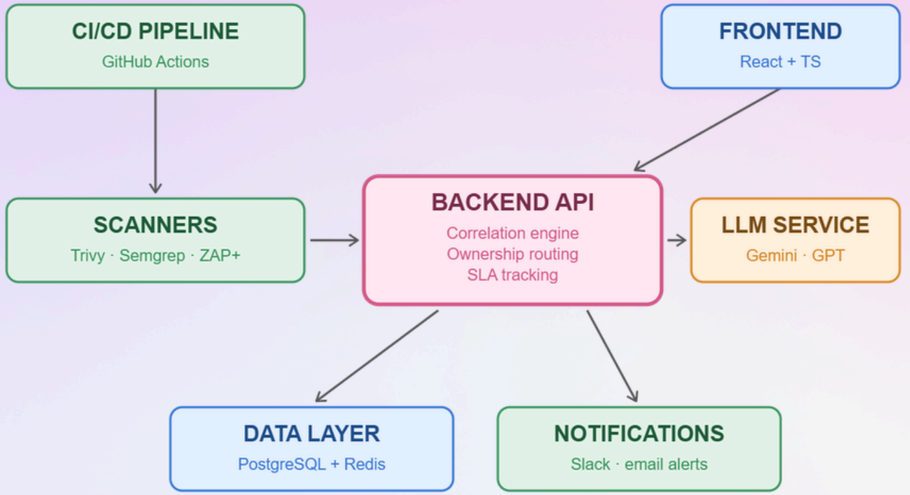
SOLUTION ARCHITECTURE

- Scanner Orchestrator — runs Trivy/Semgrep/ZAP/Gitleaks in containers
- Correlation & Dedup Engine — normalizes outputs, merges duplicate findings
- Ownership Router — assigns findings via CODEOWNERS/mapping
- LLM Service Layer — generates explanations + fixes (Gemini/GPT API)
- SLA Tracker — flags overdue findings, triggers alerts

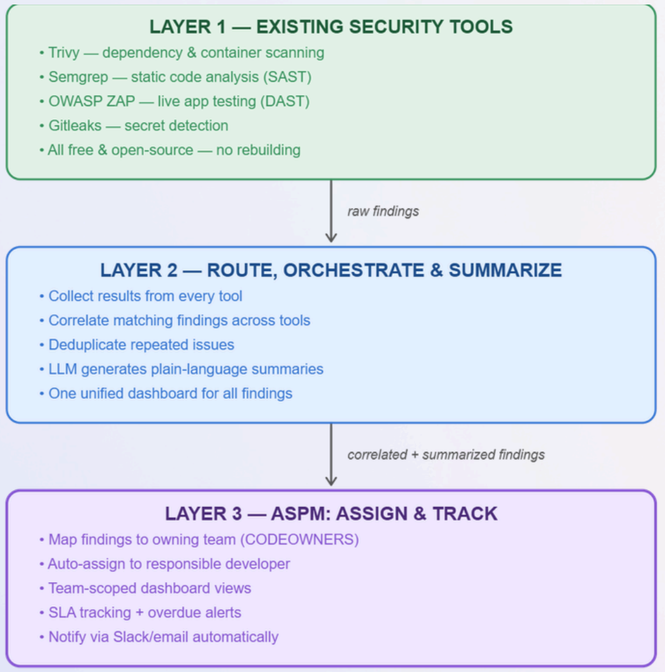


TECHNICAL ARCHITECTURE





3 LAYER ARCHITECTURE



FEASIBILITY

Technical Feasibility

All core pieces (React/TypeScript frontend, Node.js/Django backend, PostgreSQL, Redis, free scanners likeTrivy/Semgrep/ZAP/Gitleaks, LLM APIs like Gemini/GPT) are mature, well-documented technologies to use under layer 1

Economic Feasibility

Most tools are free (Trivy, ZAP, Gitleaks), LLM API costs are pennies per finding, hosting is cheap at MVP scale (~$500–2,000 total for the year, from our earlier cost breakdown). The main "cost" is your team's time, not money.

Operational Feasibility

Small teams already have GitHub/GitLab repos and CI/CD pipelines; RepoShield plugs into these via webhooks and PR comments rather than requiring a new, separate process Low friction to integrate technically.

IMPACTS & BENEFITS

IMPACT 1

Brings enterprise-grade security visibility (correlated findings, prioritized risk) to teams that previously had none

BENEFIT 1

Vulnerabilities, exposed secrets, and risky dependencies get caught and fixed earlier — before they reach production or get exploited

IMPACT 2

Removes the cost barrier that keeps small startups/student teams locked out of enterprise ASPM tools (Apiiro, Cycode, Wiz)

BENEFIT 2

Small teams get real security tooling at a price they can actually afford, instead of going without any tooling at all

IMPACT 3

Removes the manual cross-referencing of multiple scanner outputs and manual assignment of findings to developers

BENEFIT 3

Developers and security leads spend less time on tedious triage/handoff work and more time actually fixing issues faster turnaround on vulnerabilities

IMPACT 4

Building ReleaseShield exposes your team to real DevSecOps practices, API integrations, LLM application design, and security tooling

BENEFIT 4

Opens the door to a segment (compliance-sensitive small businesses) that cloud-only competitors can't easily serve

BUISNESS MODEL

TARGET USERS

- GOVERNMENT & DISASTER MANAGEMENT
- CRITICAL INFRASTRUCTURE
- WEATHER-SENSITIVE SECTORS

REVENUE STREAMS

1. GOVERNMENT / INSTITUTIONAL DEPLOYMENT
2. API-AS-A-SERVICE
3. ENTERPRISE DASHBOARDS
4. CUSTOM ANALYTICS

UNIQUE SELLING PROPOSITION

1

FAST, SELF-SERVE SETUP

Get started in under an hour with no sales calls or complex onboarding.

2

UNIFIED SECURITY VIEW

Combines Trivy, Semgrep, ZAP and Gitleaks into one centralized dashboard.

3

AI-POWERED EXPLANATIONS

Converts complex security findings into plain-language explanations and suggested fixes.

COST ANALYSIS

6-MONTH MVP1 ESTIMATE:

₹13,000 - 58,000

SCANNING TOOLS

₹0

TRIVY • ZAP • GITLEAKS • SEMGREP

FREE / OPEN-SOURCE OR FREE TIER
CORE SECURITY SCANNING WITHOUT LICENSING COST

LLM API

₹0 — MVP1

~₹500 – 1,900/MONTH POST-MVP
FREE-TIER / OPEN-SOURCE MODEL FOR MVP1
RATE-LIMITED INITIALLY; COSTS SCALE WITH USAGE

HOSTING

~₹2,000 – 9,500/MONTH

- NO KUBERNETES REQUIRED AT MVP SCALE
- LIGHTWEIGHT INFRASTRUCTURE KEEPS INITIAL COSTS LOW

AUTH + DOMAIN

AUTH: ₹0

AUTH0 / CLERK FREE TIER
DOMAIN + MISC:
~₹1,500 – 2,850/YEAR

Costs Scale After MVP

LLM and hosting costs increase with usage, while scanning tools and auth can remain free.

